

Project Title	Design and Implementation of a Highly Directional Loudspeaker System
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This Revision	V1

Scope Statement

The scope of this project involves the design, development and testing of a highly directional loudspeaker system that uses beamforming techniques to control the direction of sound. Directional loudspeakers can focus audio towards specific listeners while minimising sound in other areas. The project will focus on simulating, developing and testing a loudspeaker array that uses constructive and destructive interference to create separate sound zones and minimise audio bleed between listeners. The overall aim is to achieve clear, focused sound delivery to chosen areas while reducing sound leakage and interference between zones.

Aims

- 1) To design and implement a highly directional loudspeaker system using beamforming principles.
- 2) To develop a system capable of producing multiple focused audio zones for different listeners.
- 3) To gain a deeper understanding of how variations in signal timing, phase and amplitude affect sound directionality and clarity.

Objectives

- 1) System Design: Create a theoretical and simulated model of a loudspeaker array to analyse beamforming performance and sound direction control.
- 2) Prototype Development: Build a prototype and implement hardware and software capable of controlling signal delay, phase, and amplitude.
- 3) Testing and Evaluation: Measure the system's ability to focus sound and assess the level of sound leakage between zones. Refine the design based on test results.
- 4) Documentation: Record the full process of design, simulation, construction and testing through all required project reports and deliverables.

The primary goal is to develop a functional prototype that can reliably deliver clear, directional sound to specific areas. The project will focus on simulation, prototype construction and performance testing.

Stakeholder List

Stakeholder	Power	Interest	Concerns
Supervisor (Owner)	High	High	On time & achieves milestones/outputs Timely solutions to project issues Regular Reports & achieve deliverables
Co-Assessor	Medium	Low	Achieve deliverables
UG Ethics Committee	Medium	Low	Provides Independent Peer Review to confirm that Ethical guidelines are followed

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Deliverables

Fixed Deliverables

These are standard for all projects and should be incorporated into your Work Breakdown and Work Schedules below.

- Project Plan (working 'live' document)
- Progress Report & Presentation (Word document + PPT document)
- Project Dissemination (poster/video/web page/social media/reflective essay)
- Bench Demonstration & Interview (demonstrator)
- Final Project Report (Word document)

Interim Deliverables

- Simulation Demo
- Simulation Report

Work Breakdown	
	1) Background Research & Foundation Techniques 2) System Design & Simulation 3) Prototype Development & Core Implementation 4) Testing & Evaluation 5) Documentation & Final Deliverables

Work Package #1	Background Research & Foundation Techniques
Task #1	Background Research: Conduct a review on directional loudspeakers, phased arrays and beamforming techniques. Focus on constructive and destructive interference, multi-zone audio and methods to reduce sound leakage. Includes reviewing academic papers and practical demonstrations to understand real-world applications.
Task #2	Foundation Techniques: Investigate core directional audio principles, including phase and amplitude control, time-delay calculations and interference manipulation. Consider acoustic effects such as reflections, diffraction and side-lobes.
Task #3	Component Research: Review existing hardware technologies for loudspeaker arrays. Compare these technologies against approaches identified during the background research, assessing their suitability in beamforming and multi-zone audio applications. Look at the pros and cons of each different technology.
Linked Deliverables	Progress Presentation, Project Plan

Expected Outputs	To develop a good understanding of project from a theoretical and technical point of view.
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Work Package #2	System Design & Simulation
Task #1	Array Design: Design a loudspeaker array, consider speaker numbers, spacing and layout in array design.
Task #2	Beamforming Simulation: Develop simulation models using software, based off the array design. Visualise constructive and destructive interference patterns, sound intensity distribution and zone separation. Test multiple listener positions to analyse sound leakage and side-lobe effects.
Task #3	Parameter Optimisation: Analyse simulation results to optimise array configuration and signal parameters. Determine ideal delay, phase, and amplitude settings to achieve focused beams while minimising bleed. Evaluate trade-offs between beam width, coverage and system complexity.
Linked Deliverables	Progress Presentation, Final Report, Simulation Demo, Simulation Report
Expected Outputs	To design and simulate a loudspeaker array capable of steering sound and forming separate audio zones.

Work Package #3	Prototype Development & Core Implementation
Task #1	Hardware Assembly: Construct the loudspeaker array based on the optimised design. Ensure the physical alignment and spacing are precise to maintain accurate beamforming performance.
Task #2	Signal Control Implementation: Develop the software systems to control the timing, phase, and amplitude of signals for each speaker. Integrate with hardware.
Task #3	Multi-Zone Beam Steering: Configure the array to produce multiple steerable audio beams simultaneously. Ensure separate audio zones can carry different content with minimal interference or audio bleed between zones.
Linked Deliverables	Bench Demo, Final Report
Expected Outputs	To develop a working prototype of the loudspeaker array and demonstrate basic beamforming control.

Work Package #4	Testing & Evaluation
Task #1	Acoustic Measurement: Perform tests to measure beam directionality, sound intensity distribution and inter-zone leakage. Capture and visualise data from tests. Compare measurements with simulation predictions.
Task #2	Performance Evaluation: Evaluate prototype performance for beam accuracy, zone separation and interference suppression. Identify sources of distortion, side-lobes or unintended audio bleed.
Task #3	System Optimisation: Refine beamforming parameters, signal timing, amplitude or array geometry to improve directional accuracy and reduce reflections or interference.
Linked Deliverables	Bench Demo, Final Report
Expected Outputs	To test and refine the system to improve sound direction, clarity and zone separation.

Work Package #5	Documentation & Final Deliverables
Task #1	Project Documentation: Compile all research, design decisions, simulations, prototype data and test results into the project report. Include figures, plots, and acoustic measurements to support conclusions.
Task #2	Presentation Materials: Prepare visual aids, plots and audio demonstrations for the final presentation. Highlight prototype performance and beamforming functionality to showcase project achievements.
Linked Deliverables	Final Presentation, Final Report
Expected Outputs	To present the final system design and results.

Work Schedule

Semester 1

- Research
- Project plan
- Weekly quizzes and progress tracking
- Order and collect components
- Begin initial testing and prototype setup
- Prepare progress presentation and demo

Semester 2

- Full system implementation and integration
- Testing, debugging and performance evaluation
- Project enhancements and refinements
- Prepare final poster and demo presentation
- Complete and submit final thesis

WP #	Task #	Dependencies	Start Date	End Date	Deliverable
WP1	T1	N/A	23/09/25	20/10/25	
	T2	T2 FS T1	21/10/25	07/11/25	Project Plan
	T3	T3 FS T2	18/11/25	28/11/25	Progress Presentation
WP2	T1	T1 FS WP1 (T1 + T2)	15/10/25	28/11/25	Simulation Demo/Report, Progress Presentation
	T2	T2 FS WP1 (T1 + T2)	15/10/25	28/11/25	Simulation Demo/Report, Progress Presentation
	T3	T3 FS WP1 (T1 + T2)	15/10/25	01/12/25	Simulation Demo/Report, Progress Presentation
WP3	T1	T1 FS WP2	12/01/26	25/01/26	Bench Demo
	T2	T2 FS T1	26/01/26	23/02/26	Bench Demo
	T3	T3 FS T1	26/01/26	23/02/26	Bench Demo
WP4	T1	T1 FS WP3	16/02/26	09/03/25	Bench Demo/Final Report
	T2	T2 FS WP3	16/02/26	09/03/25	Bench Demo/Final Report
	T3	T3 FS T2	16/02/26	09/03/25	Bench Demo/Final Report
WP5	T1	T1 FS WP1-4	TBD	TBD	Final Report
	T2	T2 FS WP1-4	TBD	TBD	Final Presentation

WP #	Task #	Sept	Oct	Nov	Dec	Jan	Feb	Mar	April
WP1	T1	X	XXX						
	T2		XX	XX					
	T3			XX					
WP2	T1		XX	XXXX					
	T2		XX	XXXX					
	T3		XX	XXXX	X				
WP3	T1					XX			
	T2					X	XXX		
	T3					X	XXX		
WP4	T1						XX	XX	
	T2						XX	XX	
	T3						XX	XX	
WP5	T1							XX	XX
	T2							XX	XX

Risk Register & Risk Mitigation Plan

Risk Register

Risk	Probability (1-5)	Impact (1-5)	Priority (1-25)	Triggers	Response Plan
Hearing Hazards	1	5	5	Testing at high volume levels	Wear hearing protection if exposed to high volume. Limit exposure time and maintain safe test distances.
Electrical Hazards	1	5	5	Malfunction of electrical components, poor maintenance	Use appropriate PPE. Regularly inspect equipment for electrical safety. Immediate response to electrical malfunctions.
Soldering	1	5	5	Improper use of a soldering iron, non-use of PPE while soldering	Use tweezers/pliers or a vice to hold work piece where possible. Always assume that the soldering iron is hot and place back in its holder when not being used. Switch the iron off when it is not in use and replace in holder
Solder fumes	1	2	2	Soldering in a badly ventilated area, not using an extractor for fumes.	Solder using an extractor and always solder in a well-ventilated area.
Fire	1	5	5	Faulty equipment, poorly constructed electrical circuits or component failure may lead to short circuit, causing a spark, skin burns. Leaving hot soldering iron/workpiece in contact with combustible items may lead to fire.	Always use a damp sponge for wiping the soldering irons tips (not paper towels). A competent person e.g., supervisor/technical staff should check circuits. Visually inspect all "in-house" constructed circuits for short circuits, incorrectly orientated components e.g., capacitors, batteries etc., before connecting to a power supply.

Mitigation Planning:

Response #	Response Actions Planned	Person?	Timeframe?	Active?
1	Learn safe handling of electrical components and circuits. Use appropriate PPE. Regularly inspect equipment for electrical safety.	Health and Safety Officer	Ongoing	Yes
2	Implement a procedure for safe soldering practices, including using tweezers/pliers or a vice to hold workpieces, ensuring proper use of PPE, and incorporating safety measures for soldering fumes.	Health and Safety Officer	Ongoing	Yes
3	Conduct regular inspections of circuits for faults, ensuring proper ventilation during soldering, and establishing ergonomic measures to reduce physical discomfort during prolonged soldering.	Health and Safety Officer	Ongoing	Yes

4	Implement a checklist for visual inspections of circuits before connecting to a power supply, conduct regular training on fire safety, and establish protocols for equipment checks.	Health and Safety Officer	Ongoing	Yes
5	Implement an emergency response protocol for fire incidents, including evacuation procedures, fire extinguisher training, and regular fire drills.	Fire safety Officer	Monthly	Yes

Communications Strategy

Name	Recipient(s)	Frequency	Medium	Contents
Status Reports	Supervisor	Weekly (Friday)	SharePoint (email)	Bullet point summary of activities, blocking issues, etc.
Supervisor Meetings	Supervisor	Monthly (Friday)	Zoom/Teams (f-t-f)	Summary of progress during the period; note any blocking issues & solution or still open
Progress Report & Presentation	Supervisor & Co-Assessor	Once (deliverable)	Zoom/Teams (f-t-f)	As per guidance.
Poster	All Stakeholders	Once (deliverable)	Public Event	As per guidance